Universidade de Aveiro



universidade de aveiro
theoria poiesis praxis

SynopStudio

AI Tool for Automatic Video Summarization and Editing

Milestone 2 - Elaboration

Team Members:

David Saraiva Monteiro (125793)

Tomás Manuel Chaves Lopes (125596)

Murilo Frezzato Francisco (125487)

Lucas Morgado dos Reis (126287)

Daniel Fidalgo Duque (124880)

Supervisors:

Diogo Nuno Pereira Gomes - DETI

Rui Manuel de Assunção Raposo – DECA

30th September 2026

Contents

1. Requirements Elicitation

1.1 Requirements Gathering

In accordance with the OpenUP guidelines for the elaboration phase, the requirements elicitation step must rely on structured sources and verifiable reference materials. For the SynopStudio project, we identified the following sources:

- **Oficial Project Proposal formulated by supervisors Diogo Gomes (DETI) and Rui Raposo (DECA):** Crucial for establishing core goals, multimodal challenges and initial functional objectives.
- **Stakeholder Feedback:** Steering and alignment meetings with project supervisors Diogo Gomes and Rui Raposo.
- **Collaborative Sessions:** Internal brainstorming sessions conducted by the development team to delineate use cases and modular boundaries.
- **State-of-the-Art Benchmarking:**
    - Functional analysis of automated short-form clipping tools such as OpusClip, pivotal for identifying gaps in informational completeness, scientific rigor and context preservation.
    - Evaluation of AI-assisted video editors such as CapCut with the objective of highlighting the lack of document integration for auxiliary reference and the absence of automated user-driven specifications
    - Assessment of visual design tools like Canva to recognize their limitation to basic video editing rather than a deep semantic summarization and structured synthesis
- **Observation and Interviews of Target Users:** Requirements gathering from academic, institutional and political creators such as lecturers, researchers and political debaters. The primary focus is to extract short clips from long lectures

and political debates without compromising contextual fidelity, argumentative rigor, and editorial neutrality. Identification of workflow bottlenecks among independent content creators, podcasters and streamers struggling with the manual labor of cutting, captioning and reframing.

1.2 Functional Requirements

The functional requirements are grouped into coherent functional packages/modules, following software engineering best practices.

1.2.1 - Module 1: Project Management, Licensing and Source Ingestion

- **FR01 – Metadata, Authorship and Licensing Registration**

  Description and Inputs

  The platform must allow users to create a new project by registering mandatory metadata, including video title, authorship, speakers (people participating in the video), recording date, a license (Creative Commons, institutional permission, public political debate broadcast rights) and copyright notices.

  Processing Pipeline and Business Rules

  The system must enforce explicit confirmation of copyright ownership and usage authorization before processing can begin. Metadata must be permanently associated with the project record for institutional auditing and regulatory compliance.

  Output

  Persisted project record with a unique identifier and compliance metadata.

  Acceptance Criteria:

  The AI pipeline cannot be triggered without valid licence selection and authorship confirmation.

- **FR02 – Multi-Source Long-Form Video Ingestion**

  Description

The platform must ingest long-form video recordings such as academic lectures, conferences, seminars, political debates, streams and gameplays. This can be done through either direct file upload or authorized external streaming links, including Youtube URLs and video repository endpoints. The system must validate formats, verify media integrity, extract audio tracks and enqueue the media asset for asynchronous background processing.

Inputs

Direct File Upload: Video files in standard container formats (MP4, MOV, MKV, WebM, AVI), encoded with common video codecs (H.264/AVC, H.265/HEVC, VP9, AV1) and audio streams (AAC, MP3, Opus, PCM).

Remote URL / Youtube Link: Valid HTTP/HTTPS streaming URL pointing to an authorized online video (Youtube video/livestream, Vimeo, institutional media repository).

Unique Project Identifier generated in FR01

Optional User Constraints such as ingestion timecode boundaries (start and end times) that allow partial ingestion of long parliamentary sessions or extended lectures.

Processing Pipeline and Business Rules

1. Source Parsing and Input Routing

   Via Direct Upload, the client-side application verifies file extension and MIME type, initiating a chunked multipart upload session to ensure a robust transfer of large files up to multi-gigabyte recordings

   Via Remote URL / Youtube link, the server validates URL syntax and domain legitimacy, querying the external platform to confirm that the target resource is publicly accessible, unlisted accessible or authorized. There must be a verification that the duration falls within platform limits.
2. Media Stream Retrieval

   Direct uploads are assembled and written to secure staging object storage. For youtube and external repository links, an asynchronous ingestion worker fetches the source stream to retrieve the accurate audio track and a video stream up to 1080p resolution. This information goes directly to the storage bucket without requiring manual client downloads.
3. Container and Stream Integrity Inspection

   The system runs automated media inspection via demuxing libraries such as FFprobe, to validate file headers, detect corrupted video packets, check for variable frame rates, and verify the presence of at least one valid audio track. Silent recordings or unreadable files are rejected immediately. The

project status must then be updated to “Ingestion Failed” with explicit error details.

4. Audio Extraction and Normalization

   The audio stream is isolated and converted to an uncompressed 16 kHz, 16-bit mono PCM WAV format to optimize acoustic recognition for subsequent audio transcription models. An audio waveform and volume profile are generated to assist visual scrubbing in the editing interface.

5. Visual Proxy and Keyframe Generation

   A lightweight proxy video (720p H.264 as an example) is generated alongside an index of video keyframes to ensure responsive playback and timeline scrubbing inside the browser editor. This prevents heavy bandwidth loads during review.

6. Metadata Persistence and Asynchronous Dispatch

   Technical metadata is saved to the database (total runtime, dimensions, aspect ratio, frame rate, audio sample rate, source URL or original file name). The system marks ingestion as complete and issues an asynchronous message broker event to trigger the transcription and diarization pipeline.

Outputs

Staged raw media asset and normalized audio file stored in project storage.

Extracted media technical profile (duration, resolution, container type, codec details).

Video proxy and keyframe index for the web interface

Real-time progress notifications sent to the user interface (“Downloading from URL”, “Extracting Audio”)

Acceptance Criteria

Direct upload accepts and processes MP4, MOV, MKV amd WebM files without data corruption.

Submitting a valid Youtube URL automatically downloads and stages the media without manual user intervention.

Ingestion of video and extraction of audio execute asynchronously without blocking or timing out the user interface

Files lacking a decodable audio track or with invalid URLs are caught with descriptive error messages.

- **FR03 – Ingestion of Authorized Auxiliary Context Documents**

Description and Inputs

The platform must allow users to upload authorized auxiliary context files such as scientific papers, presentation slides, political debate briefing notes or technical glossaries to guide speech transcription accuracy and disambiguate specialized terminology. The text and presentation documents must be in PDF, PPTX, DOCX, or TXT formats.

Processing Pipeline and Business Rules

The system parses uploaded documents to extract plain text, headings, speaker bios and domain-specific acronyms. Subsequently, it builds a normalized keyword and entity lexicon associated with the project ID to directly guide speech recognition models and prompt open LLMs.

Outputs

Parsed domain vocabulary and entity index stored alongside project metadata

Acceptance Criteria

The text extracted from slides, academic papers, or other sources is successfully indexed and exposed to the downstream transcription refinement module.

1.2.2 - Module 2: Multimodal Processing and AI Transcription

- **FR04 - Synchronized Audio Transcription and Time Alignment**

Description

The platform must convert spoken dialogue into synchronized, timestamped text at both sentence and word levels, natively supporting Portuguese with an open, decoupled architecture designed to accommodate additional languages. The transcription engine must handle diverse acoustic contexts, from formal academic lectures to fast-paced political debates and interactive roundtables.

Inputs

Normalized 16 kHz mono PCM WAV audio track generated during the ingestion phase.

Language configuration parameter (default: European Portuguese, extensible to English and other languages).

Acoustic context profile tags to optimize acoustic hyperparameters. Tag examples: "Academic Lecture", "Political Debate", "Conference", "Interview".

Processing Pipeline and Business Rules

Outputs

Acceptance Criteria

- **FR05 – Speaker Identification and Voice Turn Attribution**

Description

The platform must partition the audio stream according to individual speaker identities (who spoke when), attributing speech turns across multi-participant recordings. The system must resolve complex conversational dynamics, including rapid turn-taking, distinct vocal profiles, and overlapping arguments typical of political debates and academic panels.

- **FR06 – Terminology Refinement Using Document Context**

Description

The platform must dynamically refine the transcription output by comparing ambiguous or low-confidence words against the specialized vocabulary extracted from auxiliary context documents (e.g., academic articles, presentation slides, political debate dossiers, legislative agendas). This ensures that technical terminology, political entities, and domain-specific acronyms are accurately represented.

1.2.3 - Module 3: Semantic Segmentation and Natural Language Search

- **FR07 – Structural Segmentation by Topics, Scenes and Interventions**

Description

The platform must partition long format video recordings into coherent thematic and structural units by analyzing semantic shifts, visual scene variations, and speaker transitions. This structural division applies to academic lectures, roundtables, and political debates, enabling modular content organization.

- **FR08 – Natural Language Semantic Search**

Description

The platform must allow users to query the entire video recording, transcript, and thematic chapters using free text queries in natural language, retrieving relevant video moments based on conceptual meaning rather than exact keyword matches.

1.2.4 - Module 4: Intelligent Clip Proposal and Synthesis

- **FR09 – Summarization and Output Configuration**

Description

The platform must allow the user to define operational parameters for clip synthesis, including target duration, thematic focus, target audience profile, and social media layout proportions.

- **FR10 – Automated Clip Generation Focused on Rigor**

  Description

The platform must automatically generate candidate video clips prioritizing intellectual rigor, argumentative coherence, and informational completeness, deliberately treating viral social media engagement as a secondary consideration.

- **FR11 – Verifiable Traceability and Context Loss Warnings**

  Description

The platform must preserve an unbroken, verifiable link between every proposed clip and its exact source position in the full-length recording, generating explicit warnings whenever discontinuous edits or omitted speech risk altering the speaker original meaning.

1.2.5 - Module 5: Workspace Editing and Human Editorial Control

- **FR12 – Hybrid Timeline and Synchronized Transcript Video Editing**

  Description

The platform must provide a synchronized editing workspace combining a text transcript with an interactive visual media timeline, enabling users to cut video, trim boundaries, and adjust playback through text manipulation or timeline controls. The hybrid interface allows educators, researchers, and political communication teams to edit footage without traditional video editing software complexity.

- **FR13 – Visual Confidence Scoring and Uncertainty Highlighting**

  Description

The platform must display visual confidence indicators across the transcript and timeline, highlighting words, speaker attributions, and summaries that have lower statistical certainty from underlying artificial intelligence models. This guides human attention toward technical terms in academic lectures or speaker names in political debates.

- **FR14 – Automated Visual Reframing and Multi Subject Adaptation**

  Description

The platform must adapt video compositions between horizontal sixteen to nine formats and vertical nine to sixteen formats, dynamically centering active speakers or visual slide materials for mobile social media channels.

- **FR15 – Explicit Human Approval and Hallucination Prevention**

  Description

The platform must mandate explicit human verification before final clip rendering, ensuring that subtitles match spoken dialogue verbatim and preventing artificial intelligence models from fabricating quotes or misrepresenting speakers.

1.2.6 - Module 6: Textual Synthesis, Motion Overlays and Media Rendering

- **FR16 – Artificial Intelligence Metadata Generation with Provenance Tagging**

  Description

The platform must generate suggested titles, concise descriptions, and topic tags for each approved clip, attaching clear provenance identifiers to distinguish machine generated text from verified human speech.

- **FR17 – Audiovisual Rendering with Visual Effects and Dynamic Subtitles**

  Description

The platform must render final short clips by assembling video cuts, dynamic reframing, styled synchronized subtitles, and configurable visual effects into production ready files for diverse social media channels. Supported effects include automated speaker identification lower thirds, digital emphasis zoom cuts, visual progress bars, and kinetic word highlighting.

1.2.7 - Module 7: Export Packaging and Editorial Accountability

- **FR18 – Audit Logging and Editorial Decision Tracking**

  Description

The platform must record and preserve an immutable chronological audit trail documenting every automatic recommendation made by artificial intelligence models and every manual alteration executed by the user during the editing session. This logging mechanism guarantees institutional accountability and verifies the contextual fidelity of final published media relative to the original recording.

- **FR19 – Export Packaging and Media Distribution Across Platforms**

  Description

  The platform must provide comprehensive mechanisms for rendering, packaging, and downloading approved clips, supporting preset configurations for social media distribution, batch archive downloads, synchronized independent subtitle files, and companion publishing metadata.

1.3 Context and State of the Art (SOA)

1.4 Personas

Eduardo Marques, 47 years old, Assistant Professor and Researcher

Eduardo teaches in the Department of Biology at the University of Aveiro. Besides teaching, he takes part in seminars and talks that are usually recorded and last about an hour. He would like to share his work on social media, but he has neither the time nor the video editing skills, so the recordings end up stored away and unused. He uses a lot of scientific terminology, which automatic transcription tools tend to get wrong. He usually has the slides and the papers associated with each presentation. His biggest concern is that a short excerpt, taken out of context, could give the wrong idea about his research.

Motivation: He would like to turn his presentations into short videos without spending time on editing, while being able to review and approve everything before publication, to make sure the content remains accurate.

António Oliveira, 52 years old, City Councilor

António is a councilor in a municipality of medium size. He speaks regularly at council meetings, municipal assembly sessions and public debates, which are recorded and last several hours. He would like residents to know about his contributions, but almost nobody watches the full recordings. He has no communications team and manages his social media himself, usually at the end of the day. Other speakers appear in the same recordings, and he is afraid of being accused of cutting the videos in a way that changes the meaning of what was said.

Motivation: He would like to share his contributions as short videos, with a link to the exact moment in the original recording, so that anyone can check the context.

João Teixeira Nunes, 24 years old, Streamer and Content Creator

João streams live three or four times a week, each stream lasting about three hours. To reach more people, he posts the best moments as short videos on other social networks. Currently he has to go through the recordings looking for those moments and edit them in an editing program, which takes him almost as much time as the streams themselves. He has video editing experience, but he would rather use that time to create new content. He needs videos in vertical format with subtitles.

Motivation: He would like to quickly find the best moments of each stream and export them already subtitled and in the right format for each social network.

Luciano Fernandes, 21 years old, Undergraduate Student and member of the Student Association

Luciano is in his third year and is part of the student society of his degree program, where he helps organize talks and workshops with guest speakers. The sessions are recorded with a phone or a borrowed camera, often with background noise. He has no video editing experience and has little free time between classes and coursework. He uses his laptop for almost everything. So far, the society only posts photos of the events, because nobody has the time to edit the videos.

Motivation: He would like a simple tool that suggests the most interesting moments and lets him adjust the cuts without having to learn how to use a video editor.

Paula Pinto, 42 years old, Lawyer

Paula likes to film moments with family and friends: birthdays, vacations and her children's soccer games. She records everything on her phone and accumulates long videos that she rarely watches again. She has tried editing a few, but gave up because she found the programs complicated and slow. She shares the videos only with family and her closest friends. Since children appear in the recordings, she does not want the videos to be accessible to other people.

Motivation: She would like to turn the long recordings into short videos to share with family and friends, in a simple way and without losing her weekends to editing.

1.5 Actors

Content Creator
The primary user of the system. A registered user who uploads long videos and optional context documents, sets the theme, audience, duration and format of the clips, and reviews the proposals generated by the AI. The Content Creator corrects the transcript, adjusts the cuts, explicitly approves the final version and exports it in the right format for each social network. No video editing experience is assumed.

Viewer
A person without an account who reaches the system through a link shared by a Content Creator. The Viewer can watch a clip and compare it with the exact moment of the original recording and its transcript, but cannot upload, edit or export anything. Access may be public or restricted to people invited by the Content Creator.

Administrator
The user responsible for operating the platform. The Administrator manages user accounts, configures the privacy, copyright and moderation rules, and consults the log of automatic decisions and user changes to audit how the system is being used.

External Platform
Any system outside the application that communicates with it through the documented API, such as the social networks where clips are published and the video repositories from which recordings are imported.

1.6.1 User Stories

Module 1: Project Management, Licensing and Source Ingestion

US01 (FR1): As a Content Creator, I want to create a project by registering the title, authorship, speakers, date and license of the video, so that its origin and usage permission are documented.

US02 (FR1): As an Administrator, I want no video to be processed without confirmation of authorship and license, so that the platform complies with copyright rules.

US03 (FR2): As a Content Creator, I want to upload a long video directly from my computer, so that I can work with recordings that only exist locally.

US04 (FR2): As a Content Creator, I want to import a video from an authorized link, so that I do not have to download the recording and upload it again.

US05 (FR2): As a Content Creator, I want to specify the time range to import, so that I only process the relevant part of a session that lasts several hours.

US06 (FR2): As a Content Creator, I want to follow the processing progress without the interface being blocked, so that I can do other things while I wait.

US07 (FR2): As a Content Creator, I want to receive a clear message when a video is rejected (no audio, corrupted file or invalid link), so that I know what to fix.

US08 (FR3): As a Content Creator, I want to attach slides, papers or glossaries to the project, so that the transcription gets the specialized terminology right.

Module 2: Multimodal Processing and AI Transcription

US09 (FR4): As a Content Creator, I want a transcript in Portuguese with every sentence and word linked to its moment in the video, so that I can quickly locate and check any passage.

US10 (FR5): As a Content Creator, I want the transcript to identify who is speaking at each moment, so that I can select only the contributions of one person.

US11 (FR5): As a Content Creator, I want to correct the name assigned to each speaker, so that no statement is attributed to the wrong person.

US12 (FR6): As a Content Creator, I want technical terms to be corrected based on the attached documents, so that I spend less time fixing the transcript by hand.

Module 3: Semantic Segmentation and Natural Language Search

US13 (FR7): As a Content Creator, I want to see the video divided by topics, scenes and interventions, so that I can navigate a long recording without watching all of it.

US14 (FR8): As a Content Creator, I want to search the content using natural language, so that I can find a subject without knowing the exact words that were said.

Module 4: Intelligent Clip Proposal and Synthesis

US15 (FR9): As a Content Creator, I want to set the theme, audience, duration and format before generating clips, so that the proposals fit my goal and the target social network.

US16 (FR10): As a Content Creator, I want to receive several proposals for short clips that make sense on their own, so that I do not have to look for the best moments manually.

US17 (FR10): As a Content Creator, I want to know why each clip was proposed, so that I can decide whether to trust the selection.

US18 (FR11): As a Content Creator, I want each clip to show the intervals of the original video it was taken from, so that I can check the context before approving it.

US19 (FR11): As a Content Creator, I want to be warned when a cut removes necessary context, so that I do not publish an excerpt that gives the wrong idea.

Module 5: Workspace Editing and Human Editorial Control

US20 (FR12): As a Content Creator, I want to edit the video by deleting or highlighting text in the transcript, so that I can adjust the cuts without learning how to use a video editor.

US21 (FR12): As a Content Creator, I want to adjust the start and end points, reorder or reject segments and see a preview, so that I have precise control over the result.

US22 (FR13): As a Content Creator, I want passages with low confidence to be visually highlighted, so that I can focus my review where the risk of error is highest.

US23 (FR14): As a Content Creator, I want the video to be reframed automatically while keeping the speaker in frame, so that I get a vertical video without reframing it by hand.

US24 (FR15): As a Content Creator, I want no clip to be exported without my explicit approval, so that nothing is published in my name without my review.

US25 (FR15): As a Content Creator, I want the system to prevent quotes that do not exist in the recording, so that nobody is quoted as saying something they did not say.

Module 6: Textual Synthesis, Motion Overlays and Media Rendering

US26 (FR16): As a Content Creator, I want to receive suggestions for the title, description and keywords, so that publishing is faster.

US27 (FR16): As a Content Creator, I want all text generated by AI to be labeled as such, so that I know what I need to validate.

US28 (FR17): As a Content Creator, I want clips to be rendered with synchronized subtitles, so that they can be understood without sound.

Module 7: Export Packaging and Editorial Accountability

US29 (FR19): As a Content Creator, I want to export each clip in the right format for each social network, so that I do not have to convert or resize it afterwards.

US30 (FR18): As a Content Creator, I want to consult the history of automatic decisions and of my own changes for each clip, so that I can show how the final version was reached.

US31 (FR18): As an Administrator, I want to consult the audit logs of the platform, so that I can verify compliance with the privacy and moderation rules.

Without Functional Requirements:

US32: As a Viewer, I want to open the link of a clip and see the exact moment in the original recording, so that I can confirm that the meaning was not changed.

US33: As a Content Creator, I want to share a clip only with invited people, so that private recordings are not accessible to third parties.

US34: As a Content Creator, I want to delete a project and all its associated data, so that my recordings are not kept after I am done.

US35: As an Administrator, I want to manage user accounts and configure the moderation rules, so that I can control who uses the platform and how.

1.6.2 Use Cases

Content Creator:

UC01: Create Project (US01, US02)

UC02: Upload Local Video (US03, US05, US06, US07)

UC03: Import Video from Authorized Link (US04, US05, US06, US07)

UC04: Attach Reference Documents (US08)

UC05: Transcribe Video (US09, US10, US12)

UC06: Correct Speaker Names (US11)

UC07: Browse Video by Segments (US13)

UC08: Search Content in Natural Language (US14)

UC09: Configure Clip Generation Parameters (US15)

UC10: Generate Clip Proposals (US16, US17)

UC11: Review Clip Proposal (US18, US19, US22)

UC12: Edit Clip via Transcript (US20)

UC13: Adjust and Preview Clip Segments (US21)

UC14: Reframe Clip to Vertical Format (US23)

UC15: Approve Clip for Export (US24, US25)

UC16: Generate Publication Metadata (US26, US27)

UC17: Render Clip with Subtitles (US28)

UC18: Export Clip for Social Network (US29)

UC19: Consult Clip Decision History (US30)

UC20: Share Clip with Invited People (US33)

UC21: Delete Project and Associated Data (US34)

Administrator:

UC22: Consult Audit Logs (US31)

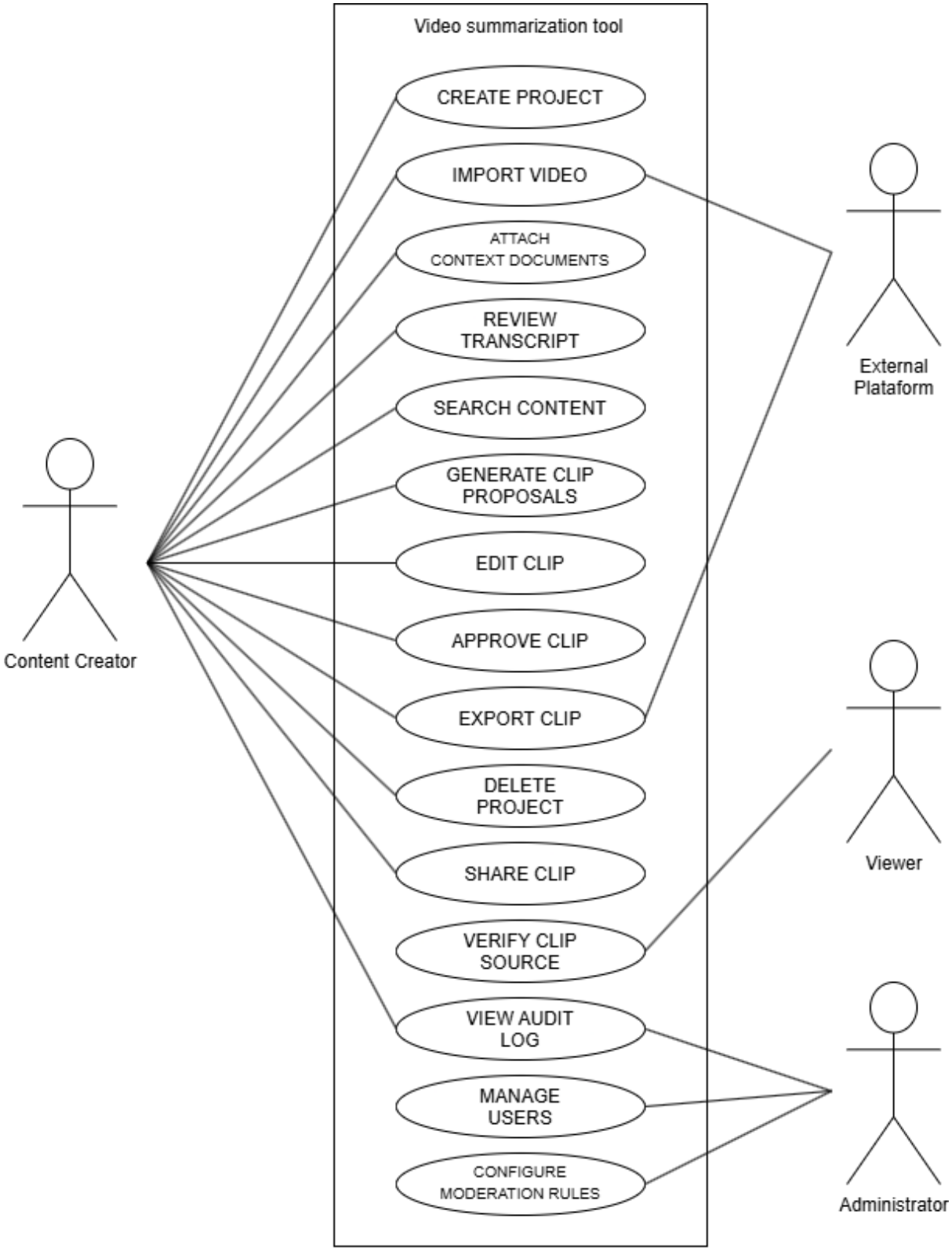
UC23: Manage User Accounts (US35)

UC24: Configure Moderation Rules (US35)

Viewer:

UC25: Verify Clip Against Original Recording (US32)

1.6.3 Use Cases Diagram



1.7 Non-Functional Requirements

Performance & Efficiency

1. The system shall process long-form videos asynchronously, so that users are not blocked while transcription, segmentation, and clip generation run, keeping UI actions under 200 ms and job submission under 5 s.
2. The system shall automate the lengthiest steps (highlight detection, cutting, captioning, reframing) to reduce manual turnaround time by at least 80%, producing publishable clips from a 1 hour recording in 15 minutes or less.
3. The system shall support efficient identification of relevant segments and speakers within long-form videos up to 4 hours, without requiring full manual review, in 5 minutes or less per hour of footage.
4. Clip generation shall support configurable durations between 5 s and 180 s (default 15–60 s), allowing the AI to propose clips within user defined length limits.

Accuracy & Reliability

1. Transcription shall be synchronized and timestamped at word level with ±100 ms accuracy, linking each phrase to its original instance in the recording.
2. The system shall support context file integration (e.g., slides, related articles) to improve transcription accuracy by at least 20% relative WER and dedicated terminology.
3. The system shall distinguish between speakers with at least 90% accuracy for up to 6 speakers, including cases where participants have similar vocal profiles, and allow manual correction when automatic identification fails.
4. Every generated clip shall be traceable to its exact source location in the original recording with ±100 ms precision, so users can verify the excerpt against the original.
5. AI-generated titles and descriptions shall not attribute to speakers statements they did not make; outputs must remain faithful to the source content, with zero fabricated quotes.

Usability & Accessibility

1. Users shall be able to review and edit video directly from the synchronized transcription, with seek latency under 300 ms and edits applied in under 1 s.

2. Users shall be able to search content by topic using natural language and select segments for clip generation, with responses under 2 s and at least 85% top 5 relevance.
3. The system shall explain or make visible why each excerpt was chosen, providing at least one rationale for 100% of clips proposed by AI, so users understand the editorial decision.

Maintainability & Modularity

1. The architecture shall be modular, allowing components (transcription, LLM, captioning, reframing) to be replaced or upgraded independently across at least 5 deployable modules.
2. The system shall expose a documented API, covering at least 95% of endpoints with OpenAPI 3.x, for integration and extension.
3. LLM modules shall be open and swappable across at least 2 providers via configuration only, avoiding lock-in to a single provider.
4. The system shall be extensible to support new platforms, aspect ratios, and languages, adding a new aspect ratio in 1 day or less and a new language in 3 days or less.

Security, Privacy & Ethics

1. The system shall preserve content integrity, ensuring 100% of published clips trace to source timestamps with zero altered quotes, so published clips reflect what was actually said (misinformation prevention).
2. The system shall handle user-uploaded videos and context files (slides, articles) securely with TLS 1.3 in transit and AES-256 at rest, enforce licence confirmation on 100% of projects, and respect copyright/ownership.
3. The system shall be transparent about AI-generated decisions (traceability + explainability), exposing rationale and source references for 100% of AI decisions and retaining audit logs for at least 12 months, to support informed editorial control.